

USMLE STEP 1 · PHARMACOLOGY

Anesthesia Pharmacology

Five-question practice set with worked explanations

PART 1 — QUESTIONS

Question 1

A 26-year-old man receives general anesthesia for an elective laparoscopic appendectomy. Minutes after induction with a volatile inhaled agent and a neuromuscular blocker, the surgeon cannot keep the abdomen insufflated because of generalized muscle rigidity. Heart rate is 132/min in sinus tachycardia and the whole body is diffusely stiff. Inhibiting which of the following would most directly reverse the underlying process?

- A. Acetylcholine breakdown at cholinergic synapses
 - B. Alpha-receptor mediated sympathetic activity
 - C. GABA-mediated hyperpolarization of neurons
 - D. Ligand binding at cholinergic (nicotinic) receptors
 - E. Calcium release from the sarcoplasmic reticulum of skeletal muscle
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Question 2

An investigational inhaled anesthetic is studied for its pharmacokinetics. In one experiment the agent is delivered at a constant partial pressure while its concentration is measured at the same time in arterial and venous blood. Early in the inhalation period there is a large difference between the arterial and venous concentrations. This finding best indicates which property of the agent?

- A. Rapid equilibration with the brain
 - B. Low solubility in peripheral tissues
 - C. Minimal uptake by peripheral tissues
 - D. Low potency
 - E. Slow onset of action
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Question 3

A 44-year-old woman with metastatic breast cancer has painful vertebral metastases that are no longer controlled by acetaminophen and NSAIDs. She is started on oral morphine. Which of the following is the most likely direct effect of this drug on the neurons of her spinal cord?

- A. Activation of the sodium-calcium exchanger
- B. Blockade of voltage-gated sodium entry
- C. Increased calcium entry into the cell
- D. Increased chloride entry into the cell
- E. Increased potassium exit from the cell

Question 4

A 45-year-old man with long-standing quadriplegia is admitted with bacterial pneumonia and, over 24 hours, deteriorates into respiratory failure requiring intubation. A skeletal muscle relaxant is given just before intubation, after which he arrests with ventricular fibrillation. A potassium level drawn during resuscitation later returns at 9.5 mEq/L. Administration of which agent is most likely responsible?

- A. Atracurium
- B. Baclofen
- C. Dantrolene
- D. Succinylcholine
- E. Vecuronium

Question 5

A 42-year-old man undergoes excisional biopsy of an enlarging pigmented forearm lesion with irregular borders. Local anesthesia is achieved with an infiltrating injection of lidocaine combined with epinephrine. What is the most likely reason epinephrine is added to the injection?

- A. It alters neurotransmission in nociceptive fibers
- B. It reduces the pain of subcutaneous infiltration
- C. It increases anesthetic efficacy through a synergistic effect
- D. It raises anesthetic potency by lowering tissue pH
- E. It prolongs the duration of action of the anesthetic

PART 2 — ANSWER KEY & EXPLANATIONS

Q1	Q2	Q3	Q4	Q5
E	E	E	D	E

Question 1 | Answer: E. Calcium release from the sarcoplasmic reticulum of skeletal muscle

This is malignant hyperthermia (MH), an inherited disorder of the skeletal muscle ryanodine receptor (RYR1). Volatile inhaled anesthetics and succinylcholine can trigger it in susceptible people. The defective receptor allows calcium to pour uncontrollably out of the sarcoplasmic reticulum into the cytosol, so muscle stays contracted. Sustained contraction consumes oxygen and generates heat and CO₂, producing early rigidity, a rising end-tidal CO₂, and tachycardia, with fever appearing later; ongoing muscle breakdown causes rhabdomyolysis and hyperkalemia. The specific antidote is dantrolene, which blocks RYR1 and shuts off the calcium release. The process that must be inhibited is therefore sarcoplasmic calcium release.

WHY THE OTHER OPTIONS ARE WRONG

- A. Blocking acetylcholine breakdown (eg, neostigmine) would strengthen, not relax, contraction.
- B. Alpha-blockers act on vascular tone and have little effect on skeletal muscle.
- C, D. Baclofen (GABA-B) and nicotinic receptor antagonists such as rocuronium act upstream of the sarcoplasmic reticulum, so neither can stop the runaway calcium release of an MH crisis.

Question 2 | Answer: E. Slow onset of action

The arteriovenous concentration gradient is the difference between how much anesthetic is in arterial versus venous blood. A large early gradient means the peripheral tissues are avidly pulling the agent out of arterial blood, that is, the drug has high solubility and uptake in tissue, leaving little in the venous return. Because so much drug is being taken up by tissue, the blood, and therefore the brain, needs longer to reach the partial pressure required for anesthesia. The result is a slow onset of action.

WHY THE OTHER OPTIONS ARE WRONG

- A. Rapid equilibration with the brain would mean fast onset, the opposite of this finding.
- B, C. Low tissue solubility and low tissue uptake both produce a small gradient and a fast onset, not a large gradient.
- D. Potency tracks lipid solubility (MAC) and is not what the arteriovenous gradient measures.

Question 3 | Answer: E. Increased potassium exit from the cell

Morphine acts on mu opioid receptors, which are coupled to inhibitory (Gi) G proteins. In the spinal cord this produces two effects. Presynaptically, voltage-gated calcium channels close, reducing calcium influx and the release of pain neurotransmitters. Postsynaptically, potassium channels open, letting potassium flow out of the cell and hyperpolarizing the neuron. That postsynaptic potassium efflux makes the second-order neuron harder to fire and dampens pain transmission, which is the direct membrane effect the question asks about.

WHY THE OTHER OPTIONS ARE WRONG

- A. The sodium-calcium exchanger is relevant to cardiac glycosides such as digoxin, not to spinal opioid analgesia.
- B. Blockade of sodium influx is the mechanism of local anesthetics such as lidocaine.
- C. Increased calcium influx is the opposite of the opioid effect.
- D. Increased chloride influx is how benzodiazepines act at GABA-A receptors.

Question 4 | Answer: D. Succinylcholine

Succinylcholine is a depolarizing neuromuscular blocker: it binds nicotinic acetylcholine receptors and holds the motor endplate depolarized. Because that receptor is a nonselective cation channel, opening it releases potassium from muscle. In patients with denervation (eg, quadriplegia, Guillain-Barre), burns, crush injury, or myopathy, nicotinic receptors are upregulated across the muscle membrane, so succinylcholine can dump a massive amount of potassium into the circulation. The result is life-threatening hyperkalemia and arrhythmia, seen here as a potassium of 9.5 mEq/L with ventricular fibrillation.

WHY THE OTHER OPTIONS ARE WRONG

- A, E. Atracurium and vecuronium are nondepolarizing blockers and are in fact the safer choice in these high-risk patients.
- B. Baclofen is a GABA-B agonist used for spasticity, not a paralytic for intubation.
- C. Dantrolene blocks RYR1 in malignant hyperthermia and does not cause this hyperkalemia.

Question 5 | Answer: E. It prolongs the duration of action of the anesthetic

Epinephrine is added to lidocaine as an alpha-1 agonist that constricts the local blood vessels. This vasoconstriction keeps the anesthetic at the injection site longer, prolonging its duration of action. It also lowers systemic absorption of the lidocaine (reducing the risk of toxicity such as arrhythmia or seizure) and decreases bleeding in the surgical field.

WHY THE OTHER OPTIONS ARE WRONG

- A. Epinephrine does not itself change transmission in nociceptive fibers; glutamate and substance P carry peripheral pain signals.
- B. It does not reduce injection pain; the acidic solution can actually sting more.
- C. It is not synergistic anesthesia, since epinephrine produces no anesthetic effect on its own.
- D. Potency is increased by raising tissue pH (eg, adding bicarbonate), not by lowering it.